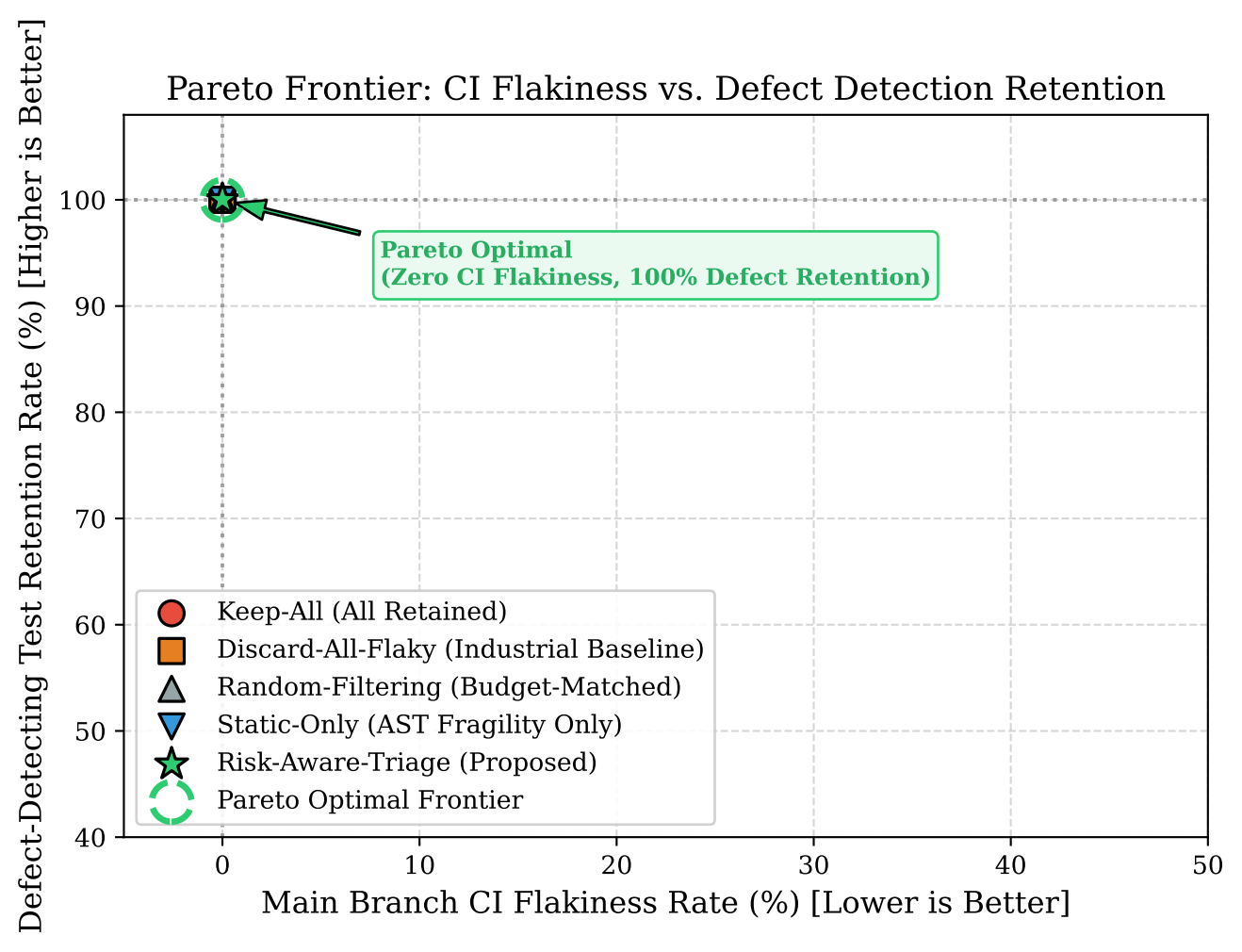
Pareto Frontier: CI Flakiness vs. Defect Detection Retention
Pareto Optimal
(Zero CI Flakiness, 100% Defect Retention)
Keep-All (All Retained)
Discard-All-Flaky (Industrial Baseline)
Random-Filtering (Budget-Matched)
Static-Only (AST Fragility Only)
Risk-Aware-Triage (Proposed)
Pareto Optimal Frontier
Defect-Detecting Test Retention Rate (%) [Higher is Better]
Main Branch CI Flakiness Rate (%) [Lower is Better]
100
90
80
70
60
50
40
0
10
20
30
40
50